BIDMC ICU Pharmacy Analytics Data Warehouse

B9DA111 Assessment ONE — Proof of Concept Data Warehouse

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Project Overview & Dataset

About the Dataset

MIMIC-III Clinical Database Demo v1.4 from MIT PhysioNet — Beth Israel Deaconess Medical Center (BIDMC), Boston.

Published by MIT with peer-reviewed documentation (Johnson et al., 2016). Genuinely transactional operational clinical data — not aggregated — with rich interconnected data across prescriptions, diagnoses, procedures, ICU stays and DRG billing.

> **Business Vision:** Consolidate ICU pharmacy data to enable BIDMC hospital leadership to analyse drug prescribing patterns across patient clinical pathways.

Key Findings

82% Both Pathway

ICU patients undergo both procedures and prescriptions, averaging **122 drugs per patient**

56% HIGH RISK

Polypharmacy patients with more than **20 distinct drugs** requiring immediate clinical review

Septicemia NOS

Most common diagnosis — **953 prescriptions** across 8 patients

Burkitt Lymphoma

Single patient outlier — **784 prescriptions** due to intensive chemotherapy

4 Stakeholders: CEO | Clinical Governance | Medical Director | Operations Manager

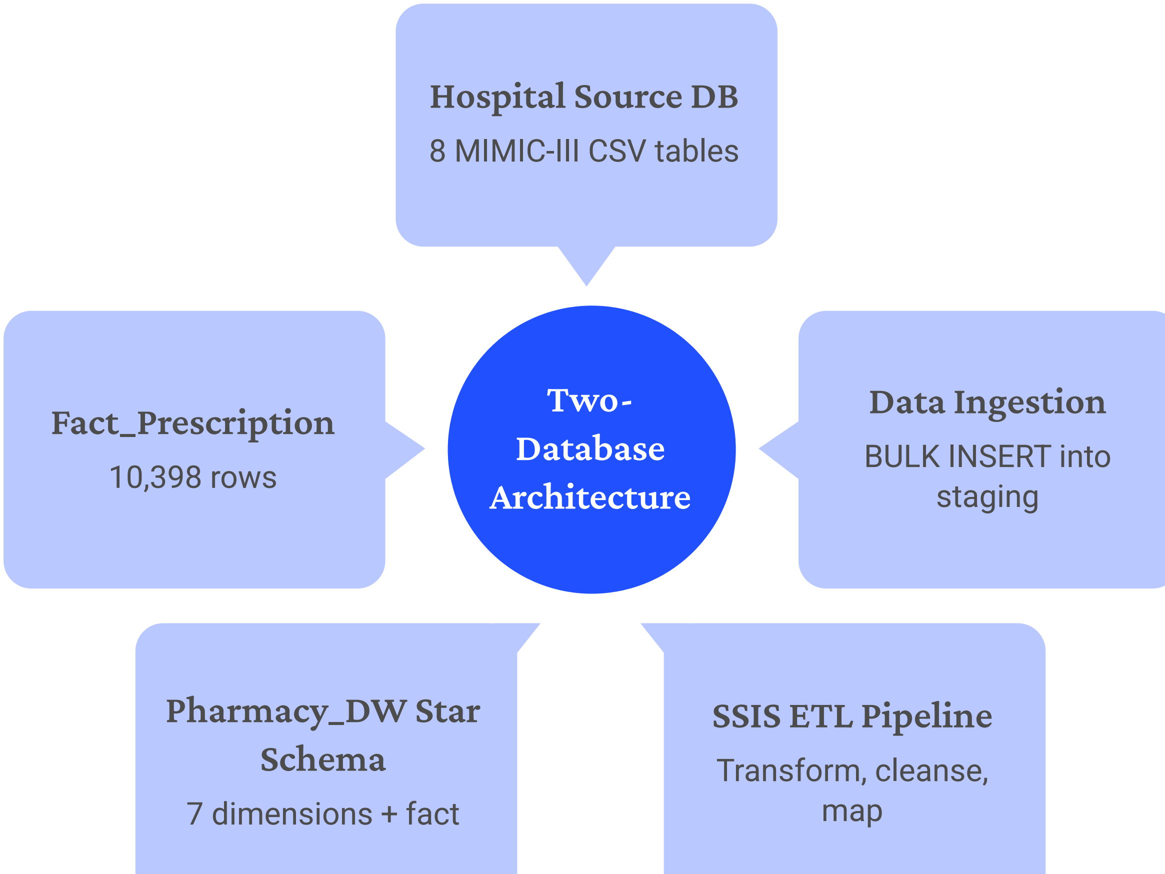
100
ICU Patients

8
CSV Source Files

10,398
Prescriptions

Data Warehouse Star Schema Design

Two-database architecture following **Kimball (2013)**: Hospital Source Database (8 tables via BULK INSERT) and **Pharmacy_DW** star schema with 7 dimensions and 1 fact table.



Star Schema Tables

Fact_Prescription

10,398 rows — atomic grain, one row per prescription event

Dim_Date

129 rows

Dim_Patient

100 rows — SCD Type 2 with ClinicalPathway derived column

Dim_Drug

592 rows

Dim_ICUStay

137 rows

Dim_Procedure / Dim_Diagnosis / Dim_DRG

165 | 582 | 161 rows

SCD Type 2

Dim_Patient tracks changes over time

Unknown Member Pattern

Key = -1 on 5 optional dimensions preserving all 10,398 fact rows

ClinicalPathway

Both 82% | Prescriptions Only 12% | Procedures Only 6%

Atomic Grain

One row per prescription event for maximum analytical flexibility

SSIS ETL Pipeline

4 SSIS Packages in **Visual Studio 2022** using ADO.NET connections — fully automated pipeline completing in **45 seconds**.

1

Package 00

Truncate DW: Drop FK constraints, truncate 8 tables, recreate FKs, insert unknown members

2

Package 01

Load Dimensions: 7 sequential Data Flow Tasks loading all dimension tables

3

Package 02

Load Fact: 10,398 rows with 6 Lookup transforms using Ignore Failure and Replace NULL Derived Columns

4

Package 03

Master Package: Single execution refreshes the entire data warehouse in 45 seconds

Key Transformations

- **DateKey:** YEAR × 10000 + MONTH × 100 + DAY
- **ClinicalPathway:** CASE statement with correlated subqueries
- **IsIVRoute:** BIT flag for IV route types
- **Unknown member expression:** `ISNULL(key) ? -1 : key`

Key Difficulties Resolved

Negative Duration: Prescription duration of -17 days resolved by replacing with 0 in Derived Column transform.

Data Type Mismatches: Required Data Conversion transforms before Lookup components to ensure type compatibility.

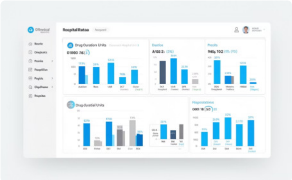
ETL Results: All 8 tables PASSED. Fact_Prescription 10,398 rows confirmed.

SSRS Reports

4 SSRS Reports developed in **Visual Studio 2022**, each tailored to a distinct hospital stakeholder with purpose-built KPIs and conditional formatting.

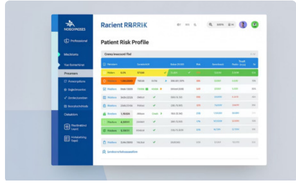
Report 1 — Executive Summary Dashboard (CEO)

6 KPI cards: Total Prescriptions **1,552** | IV Rate **54.7%** | Avg Duration **2.1 days** | Patients **100** | Formulary **99.99%** | Distinct Drugs **572**



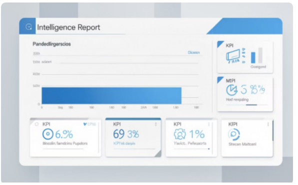
Report 2 — Prescription Duration Analysis (Clinical Governance)

MAIN, BASE, ADDITIVE drug types across 5 ICU care units. CCU MAIN drugs average **4.49 days** vs MICU **2.64 days**. Conditional formatting highlights durations over 3 days in red.



Report 3 — Patient Drug Burden & Safety Risk Profile (Medical Director)

HIGH RISK **56 patients (56%)** | MODERATE **29** | NORMAL **16**. Sorted by ICU length of stay. Most prescribed drug: Potassium Chloride (529 times).



Report 4 — Diagnosis Drug Burden Intelligence (Operations Manager)

Top 15 primary diagnoses by prescription volume. Novel KPI cards: DRGMortality **72.74%** extreme mortality risk. Burkitt Lymphoma: **784 prescriptions** for one patient.

Tableau Visualisations & Dashboard

4 Tableau visualisations connected **live to Pharmacy_DW**, combined into an interactive dashboard with filter controls for ICU Care Unit, Drug Type and Risk Level.

1	2	3	4
Drug Route Distribution by Care Unit Stacked Bar Chart — MICU and CCU show IV rates above 60%. CSRU shows more balanced IV-to-oral ratio. Supports Clinical Governance stakeholder.	**Top 10 Drugs** Horizontal bar char—D5W, 0.9% Sodium Chloride, and NS are the most used drugs, so the medication load is concentrated in a few supportive IV drugs.	**Diagnosis Complexity** Tree Map—Septicemia NOS and Brkt tmr unsp xtrndl org are the biggest blocks, so these diagnoses contribute the most to prescribing burden.	**Medication Route Share** Donut chart—Intravenous medicines take up the largest share, while oral and other routes are much smaller.

56%

HIGH RISK

Polypharmacy patients requiring immediate clinical pharmacy review

59.5%

IV Dependency

Of all prescriptions administered intravenously across ICU units

82%

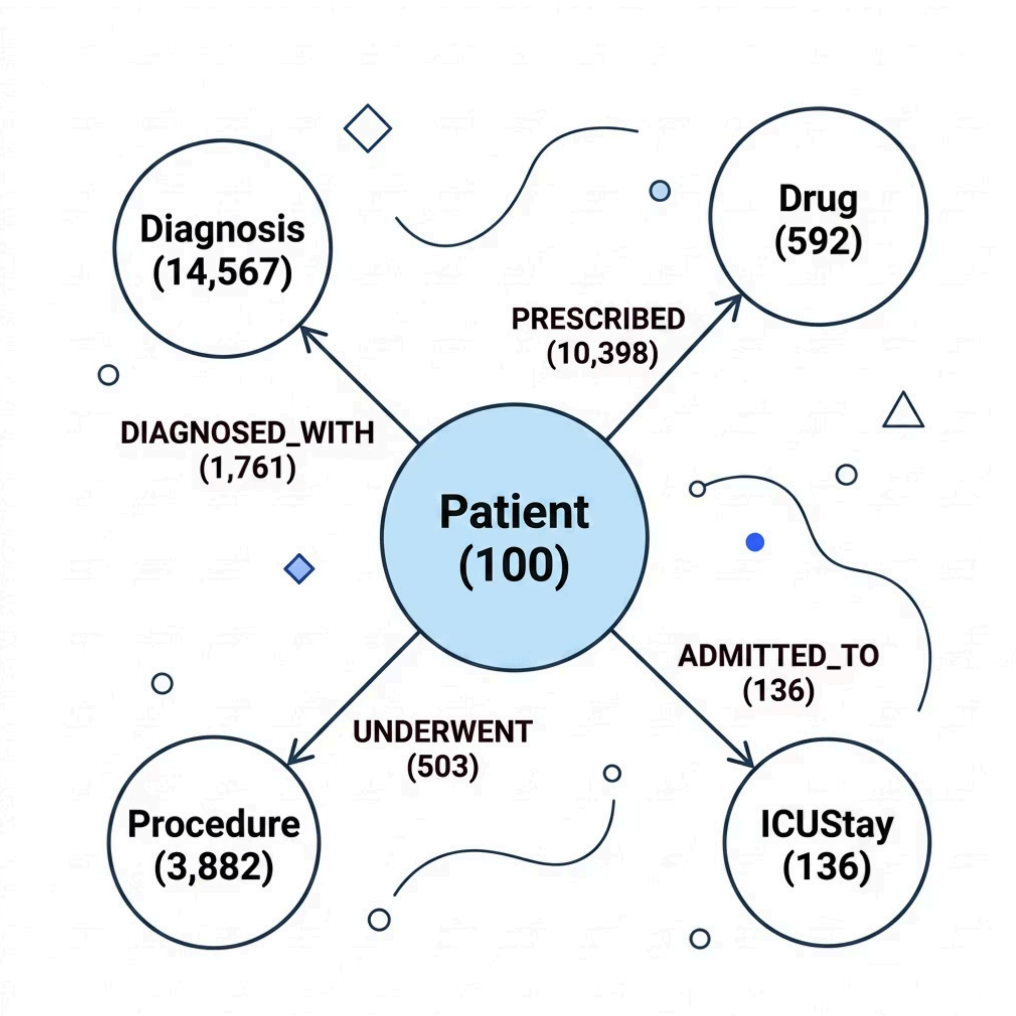
Both Pathway

Patients undergoing both procedures and prescriptions

Neo4j Graph Database Comparison

Graph schema loaded from MIMIC-III CSV files — **7 paired SQL and CQL queries** comparing relational and graph database approaches for clinical analytics.

Graph Schema



Key Difficulties & Learnings

Query 4

The Neo4j query initially inflated results because multiple MATCH clauses multiplied ICU stay and prescription rows. Although a WITH clause reduced the Cartesian product effect, the counts were still higher than SQL Server because Neo4j worked at patient level, while SQL Server used the prescription-event grain through FactPrescription and ICUStayKey. This shows the difference was structural, not just a query issue.

Query 6 & 7

The remaining mismatch happened because SQL applied **admission-level filtering** through the diagnosis join, while Neo4j counted prescriptions at the broader **patient level**. So SQL was more specific, but Neo4j included extra prescriptions from other admissions for the same patient.

Conclusion: SQL Server for structured BI reporting and SSRS dashboards. Neo4j for clinical pathway traversal and network analysis. **Hybrid architecture recommended** for production BIDMC system.

SQL Server Strength

Verbose but precise — enforces admission-level scoping through FK joins. Ideal for structured BI and SSRS dashboards.

Neo4j Strength

Intuitive pattern matching, no JOIN syntax needed. Query 2 polypharmacy: **4 lines CQL vs 10 lines SQL**. Ideal for clinical pathway traversal.

Hybrid Architecture

Recommended for production BIDMC system — combining relational precision with graph flexibility for complete clinical intelligence.

Key Insights & Clinical Findings

The Pharmacy_DW data warehouse surfaces **8 critical clinical findings** with direct implications for BIDMC hospital leadership and patient safety.

1 **Polypharmacy Risk**

56% of ICU patients are HIGH RISK requiring immediate clinical pharmacy review — more than 20 distinct drugs per patient.

2 **Procedural Complexity Drives Drug Burden**

Both pathway patients receive **3.6× more drugs** than Prescriptions Only patients (122 vs 34 average drugs).

3 **Septicemia Dominates ICU Caseload**

953 prescriptions across 8 patients — the most common primary diagnosis in the ICU cohort.

4 **IV Dependency & CCU Over-Prescribing Risk**

59.5% of all prescriptions administered intravenously. CCU MAIN drug average duration **4.49 days** vs 2.64 days in MICU.

5 **DRG Extreme Mortality**

72.74% of prescriptions associated with extreme mortality risk DRG codes — with **99.99% formulary compliance** maintained throughout.

Summary & Conclusion

A complete end-to-end proof of concept data warehouse delivering clinical pharmacy intelligence for BIDMC ICU leadership.

What We Built

01

Two-Database Architecture

Hospital source and Pharmacy_DW star schema with 8 tables, 7 dimensions and Fact_Prescription at atomic grain

02

4 SSIS Packages

Fully automated ETL pipeline with unknown member pattern, SCD Type 2 and 45-second full refresh

03

4 SSRS Reports

Purpose-built for CEO, Clinical Governance, Medical Director and Operations Manager

04

4 Tableau Visualisations

Interactive dashboard with live connection to Pharmacy_DW and filter controls

05

Neo4j Graph Database

5 node types, 4 relationship types and 7 paired SQL vs CQL queries with comparative analysis

Individual Contributions

Abin Raju

SSMS database design, SSIS ETL pipeline, Neo4j graph database and all 7 CQL queries

Keerthana Chand Mini

All 4 SSRS reports, dataset design, KPI development and clinical analysis

Balakrishna Rajesh

All 4 Tableau visualisations, dashboard design and dataset research

Technologies Used

SQL SERVER MANAGEMENT STUDIO | VISUAL STUDIO 2022 | SSIS | SSRS | TABLEAU | NEO4J DESKTOP

Key Achievement: 100% referential integrity across 10,398 fact rows using Kimball unknown member pattern. Novel ClinicalPathway derived dimension revealing 82% Both pathway patient distribution.

Thank you — Ready for Q&A and live demonstration.